Technology Stack

Date	15 March 2026
Team ID	xxxxxx
Project Name	OPTICROP – SMART AGRICULTURAL PRODUCTION OPTIMIZATION ENGINE
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your

S.No	**Architecture Component / Layer**	**Technology Chosen**	**Justification / Purpose**
1	**Frontend / Client-Side**	HTML5, CSS3, JavaScript	Used to create a responsive and user-friendly web interface where users can enter soil and weather parameters and view crop recommendations.

2	**Backend / Server-Side**	Python, Flask	Flask acts as a lightweight web framework that connects the user interface with the machine learning model and handles prediction requests efficiently.
3	**Database / Data Storage**	CSV Dataset (Crop_recommendation.csv)	Stores agricultural data such as soil nutrients, weather conditions, and crop labels used for model training and prediction.
4	**Cloud / Hosting / Deployment**	Local Flask Server (localhost)	Used for development, testing, and demonstration of the application without requiring cloud infrastructure.
5	**Version Control & CI/CD**	Git & GitHub	Helps manage source code versions, track changes, collaborate among team members, and maintain the project repository.

6	**Third-Party APIs / Other Tools**	Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib, VS Code	Scikit-Learn is used for machine learning, Pandas and NumPy for data processing, Matplotlib and Seaborn for visualization, Joblib for model storage, and VS Code for development.